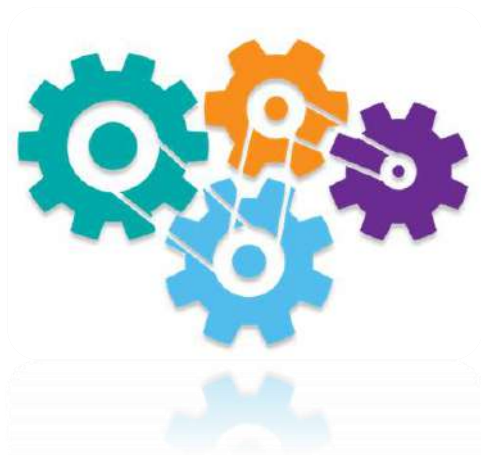




**Sutton
Ambulance
Station, Dorset
Road, Sutton,
London SM2 6JH**

Energy Strategy Report

**November
2025**



Ref: 25-15890

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The signatories below verify that this document has been prepared in accordance with our quality control requirements. These procedures do not affect the content and views expressed by the originator.

This document must only be treated as a draft unless it has been signed by the originators and approved by a director.

Revision	Initial	Rev A	Rev B	Rev C
Date:	24/11/2025			
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1. Executive Summary

This Energy Strategy Report demonstrates the predicted energy performance and carbon emissions of the proposed development at **Sutton Ambulance Station, Dorset Road, Sutton, London SM2 6JH**. The figures presented in the study are based on the most updated information provided by the project team. The proposed development is to **deliver flexible space for community, interfaith, and educational use**. The project is proposed to have a gross floor area of approximately 357 m².

The overall analysis took into consideration the latest national building regulations (i.e. **Part L2 2021**) and the local policy requirements (i.e. **London Borough of Sutton**). Based on the data, the project demonstrates compliance with the council local policies and building regulations.

1.1 Policy Requirements

Compliance with the national building regulations can be accomplished by demonstrating that the project carbon emissions are below the regulated target. In addition to the above, the local council – **Sutton Council** – requires new developments to incorporate certain sustainable design and construction measures. The table below summarises the best practice building regulations and local policy requirements the assessment adopted for the development.

Table 1: National and Local Policy Requirements

Policy	Requirement	Compliance
Part L2A	The building primary energy rate (BPER) and building emission rate (BER) must not exceed the target primary energy rate (TPER) and the target emission rate (TER), respectively.	The project achieved the criteria by improving fabric thermal performance and implementing efficient building services.
GLA Energy Sutton	As the building falls within minor developments, no need to meet the requirements set out in the London Plan.	
Sutton Local Plan 2018	As the building falls within minor developments, no need to comply GLA and to completely offset regulated carbon by renewables.	

1.2 Assessment Methodology and Strategies

The methodology adopted to mitigate the CO₂ emissions for the development is in alignment with the London Plan's Energy Hierarchy best practice. The assessment has been carried out using SBEM method which complies with the National Calculation Methodology (NCM).

Table 2 below explains the Energy Hierarchy stages and the suggested strategies proposed to help the development achieve the required targets.

Table 2: Energy Hierarchy and suggested strategies

Stages	Strategies
BE LEAN Minimising energy demand through Energy Efficient Design	- Building fabric U-values and air permeability improved beyond Part L2 requirements. - Natural ventilation with extract fans in wet rooms - Energy efficient light fittings
BE CLEAN Availability of CHP or communal heating systems	Opportunities for local CHPs and communal heating systems assessed. Further information can be found in the Be Clean Section.
BE GREEN On-site renewable technologies (i.e. ASHP, PVs, etc)	Provision of efficient building services through the use of Air Source Heat Pump. Further information to be found in the Be Green section.
BE SEEN In-use monitoring	Installation of Smart Meters is recommended to monitor the actual operational energy use, manage it effectively and mitigate the performance gap.

1.3 Assessment Results

The building has been simulated under four conditions to analyse the impact of the Energy Hierarchy factors on the results. The first simulation assesses the building under the notional building specifications, as defined within Part L. The function of this first simulation is to generate the Target Emission Rate (TER) and the Target Primary Energy Rate (TPER). The second calculation analyses the carbon reductions achieved after improving the building fabric thermal performance by adopting improved thermal U-values and passive energy efficiency measures (i.e. Be Lean). The third stage of simulation assesses the effectiveness of adopting local low-carbon energy sources like CHPs (i.e. Be Clean stage). The fourth simulation analyses the carbon reductions after considering renewable technologies (i.e. Be Green). The combined strategies result in effective design measures to achieve the >35% reductions beyond Part L2 requirements, as stipulated by the London Plan.

The project has been assessed according to the latest guidelines i.e. **Part L 2021**. As the national building regulations have been revised and updated, the compliance criteria has become significantly more demanding. As such, the development fails to meet the condition that the Building Emission Rate (BER) and Building Primary Energy Rate (BPER) remain below the Target Emission Rate (TER) and the Target Primary Energy Rate (TPER) at each stage of assessment. However, the recommended reductions over Part L baseline at Be Lean and Be Green stages are being met.

After the application of all strategies based on the Energy Hierarchy, the projected carbon emission and primary energy figures have been summarized in Table 3 below:

Table 3: Emission Rate and Primary Energy after each stage of the proposed strategy

BER (Building Emission Rate) (KgCO ₂ /m ² .annum)	Part L 2021
Baseline TER	7.42

Baseline BER	16.93
Be-Lean BER	5.73
Be-Clean BER	5.73
Be-Green BER	4.06
BPER (Building Primary Energy Rate) (kWh/m ² .annum)	Part L 2021
Baseline TPER	80.12
Baseline BPER	181.26
Be-Lean BPER	62.20
Be-Clean BPER	63.20
Be-Green BPER	43.20

Results of the simulations show that the development is able to meet the overall criteria of the national and local policies (i.e. >35% reduction over Part L 2021 requirements). Since the targeted carbon saving is a 100% reduction over the Building Regulations Part L baseline (net-zero carbon) and the building is able to achieve a 38.5% reduction through implemented strategies, the shortfall needs to be compensated through a carbon offset payment. The one-off payment has been calculated in accordance with the GLA guidance in the Table 4 below:

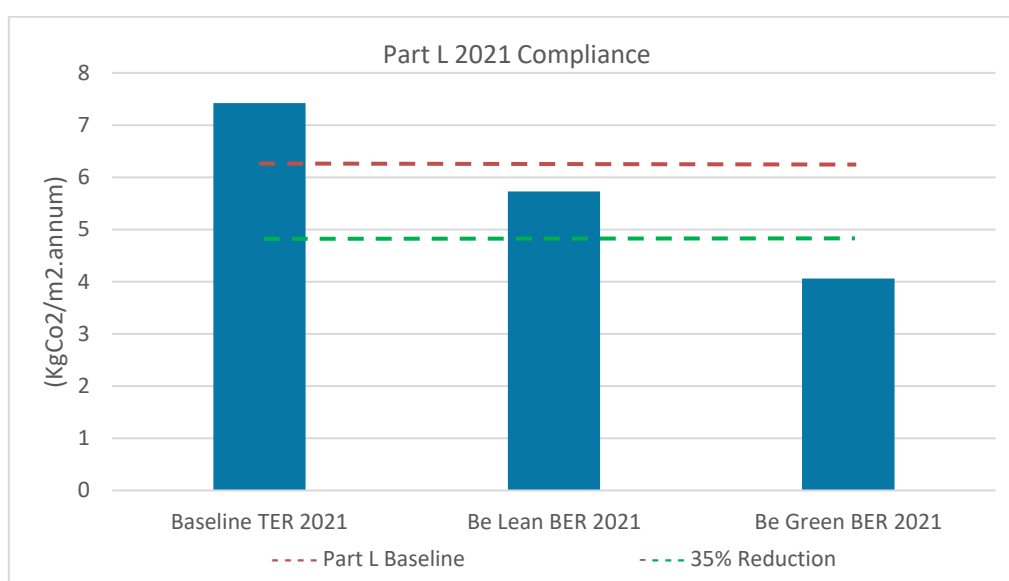


Figure 1: Part L compliance and comparison

Table 4 : Carbon offset calculation

CARBON SHORTFALL		
BER (Building Emission Rate) (KgCO ₂ /m ² .annum)	Part L 2021	% Reduction from Baseline
Baseline TER	7.42	

Be-Lean BER	5.73	22.77%
Be-Clean BER	5.73	22.77%
Be-Green BER	4.06	45.28%
CARBON OFFSET CALCULATION		
Carbon Shortfall ($\text{KgCO}_2/\text{m}^2.\text{annum}$)	4.06	
Building Area (m^2)	360	
Total Carbon to be offset (Tonnes CO_2/yr)	1.46	
Carbon offset price (£)	2850*	
Total Compensation (£)	4165.6 (One off Payment)	

* GLA recommended rate of £95 per tonne CO_2 over 30 years

(Source: Carbon Offset Funds - Greater London Authority guidance for London's Local Planning Authorities on establishing carbon offset funds)

2. Introduction

This Energy Strategy Report (ESR) focuses on the strategies studied and selected for the proposed development to meet compliance with local and national regulations. The report defines how energy consumption and carbon emissions will be minimised to meet the targeted carbon emissions stipulated as per Part L 2021. Furthermore, the selected strategies also help attain the 35% reduction over Part L, as demanded by the London Plan and local planning policy.

The assessed development is located at **Sutton Ambulance Station, Dorset Road, Sutton, London SM2 6JH** in the **London Borough of Sutton**. The development proposal has a **total built up area of 347 m²**, is for a **change of use from sui generis [ambulance station to community centre use with prayer facility]**.



Figure 2 : Site Location (as marked in orange)

3. Planning Policy

3.1 National Planning Policy Framework (NPPF, February 2021)

The National Planning Policy Framework is a key part of our reforms to make the planning system less complex and more accessible, to protect the environment and to promote sustainable growth.

3.2 The London Plan (March 2021)

Minimising greenhouse gas emissions

The London Plan is an integrated policy framework for any development being proposed in the Greater London area. It is not required for projects to exceed the national regulatory targets, meaning a minimum on-site reduction of at least 35% beyond the Building Regulations, as the development is considered minor. For minor development, There is no requirement for residential developments to achieve a 10% energy efficiency improvement, nor for non-residential developments to achieve a 15% improvement through energy efficiency measures.

The framework proposes that major developments should reduce the overall embodied greenhouse gas emissions and operational carbon emissions in accordance with the following energy hierarchy:

- A. **Be Lean:** make buildings use less energy and manage demand during operational phases.
- B. **Be Clean:** exploit local energy supply resources (i.e., District and communal heating) to deliver energy efficiently.
- C. **Be Green:** exploring opportunities for renewable energy resources by producing, storing and using renewable energy on-site.
- D. **Be Seen:** monitor, verify and report on energy performance (Part L Criteria 4 and 5).

Further guidance within the framework states that proposals should include a detailed energy strategy to demonstrate how the zero-carbon target will be met within the framework of the energy hierarchy.

3.3 London Borough of Sutton

Sutton Local Plan 2018



London Borough of Sutton
Sutton Local Plan 2016-2031

Draft Local Plan Appendix

Proposed Submission Consultation
(Regulation 19)

December 2016



Figure 3: Sutton Local Plan 2018

The Sutton Council promotes zero carbon development through its planning policy. It sets out regulation in synergy with the London Plan for all development to reduce on-site energy demand and maximise the use of low carbon energy sources to reduce the effects of climate change. Some of the guidelines specific to built environment are as below:

- A. **Carbon reduction:** A minor development is not required to be net zero carbon and does not need to demonstrate, through an energy strategy, how such a target would be achieved.

- B. **Heat networks:** For minor developments, there is no requirement to connect to existing or planned local heat networks, nor to establish a new heat network, even where this would otherwise be feasible.
- C. **Sustainable design:** Development should enable the extended lifetime of buildings and spaces and respond to the likely risks and consequences of climate change by incorporating principles of sustainable design. This can be done by demonstrating compliance with BREEAM “Excellent” or equivalent standard.

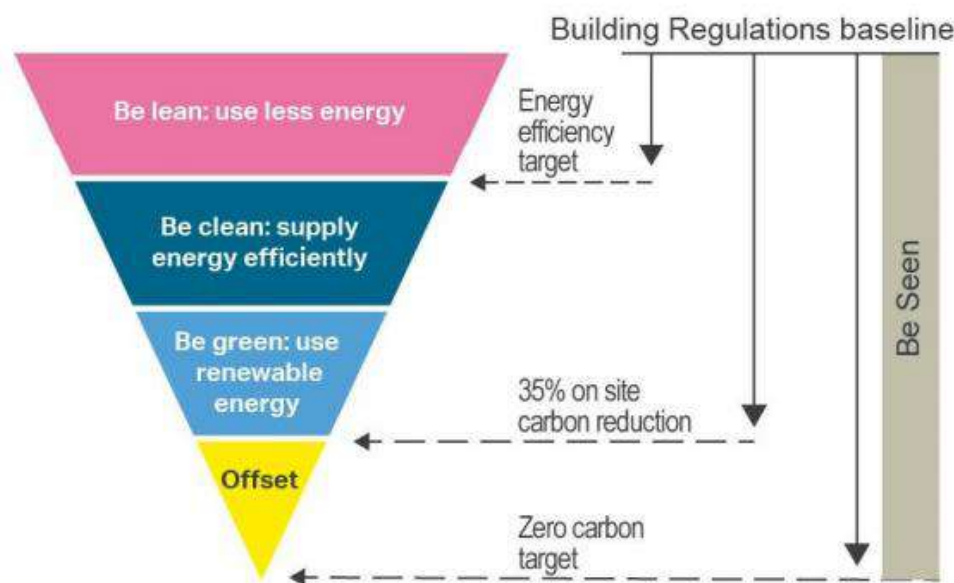
4. Assessment Methodology

4.1 The Energy Hierarchy

The energy hierarchy is a classification of different methods to improve energy performance in a parallel sequence. This primarily includes a focus on reducing energy use by avoiding unnecessary consumption, to then improving the efficiency of energy systems to minimise loss. This is followed by exploiting renewable energy sources and low carbon energy solutions for energy needs. Finally, any remaining demand can be catered for by conventional fuel sources and carbon offsetting solutions.

The Energy Strategy adopts a set of principles to guide design development and decisions regarding energy, balanced with the need to optimise environmental and economic benefits. The following hierarchy should be used to assess applications:

- **BE LEAN** – By using less energy and considering the further energy efficiency measure in comparison to the baseline building.
- **BE CLEAN** – By supplying energy efficiently. Clean energy use looks at further carbon dioxide emission savings over the lean building by taking into consideration the use of decentralised energy (e.g. CHP, District Heat Networks).
- **BE GREEN** – By integrating renewable energy into the scheme which can further reduce the carbon dioxide emission rate.
- **BE SEEN** – By monitoring, verifying, and reporting on energy performance to use energy mode effectively.



Source: Greater London Authority

Figure 4: The Energy Hierarchy

4.2 Modelling strategy

The Government approved software; IES-VE has been utilized to carry out the project compliance simulations (SBEM) according to the National Calculation Methodology (NCM). Simulated models are built to assess the actual building BER against the notional building TER. The notional building used to determine carbon emission target (TER) is the same size and shape as the actual buildings, constructed to concurrent regulated specifications (i.e., **Part L2 2021**).

The actual building has been modelled entirely to the notional building specifications in order to meet the carbon targets and the limiting fabric and buildings services parameters. However, for differences in fabric design and glazing areas, actual buildings sometimes are expected to exceed the notional TER. Therefore, further improvements to the actual building parameters (e.g., fabric, HVAC, lightings, renewables) are made to meet the required compliance targets. The approved document (Part L2) however, encourages developers to vary the specification provided the same overall level of carbon dioxide emissions is achieved or bettered. It is important to note that SBEM is not intended to be used as a building design tool but to inform design decisions.

Syntegra received architectural drawings and project relevant documents. Received information is used to undertake the ultimate energy assessments and supporting the modelling assumptions. The document references are listed on the table below.

Table 5: Reference document list

No.	Document Name	Format.	Received Date
1	786-Basma-Pp-200-Proposed Site Layout Plan, Site Location Plan & Block Plan	Pdf	30 -05-2025
2	786-Basma-Pp-201-Proposed Ground Floor Plan, Birds-Eye Roof Plan And 3d View	Pdf	30 -05-2025
3	786-Basma-Pp-202 - Material Elevation A - May 2025	Pdf	30 -05-2025
4	786-Basma-Pp-203 - Material Elevation B - May 2025	Pdf	30 -05-2025
5	786-Basma-Pp-204-Proposed Sections	Pdf	30 -05-2025
6	786-Basma-Su-100 - Existing Ground Floor and Bird-Eye Roof Plan, Site Location Plan and Block Plan	Pdf	30 -05-2025
7	786-Basma-Su-101-Existing North, West, South & East Elevations and Sections	Pdf	30 -05-2025
8	786-Basma-Su-102-Existing Building 3d Views	Pdf	30 -05-2025

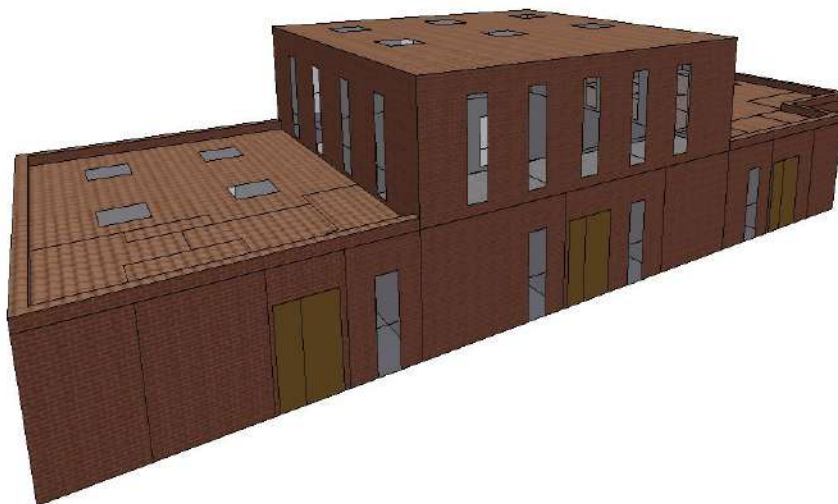


Figure 5: View of the energy model

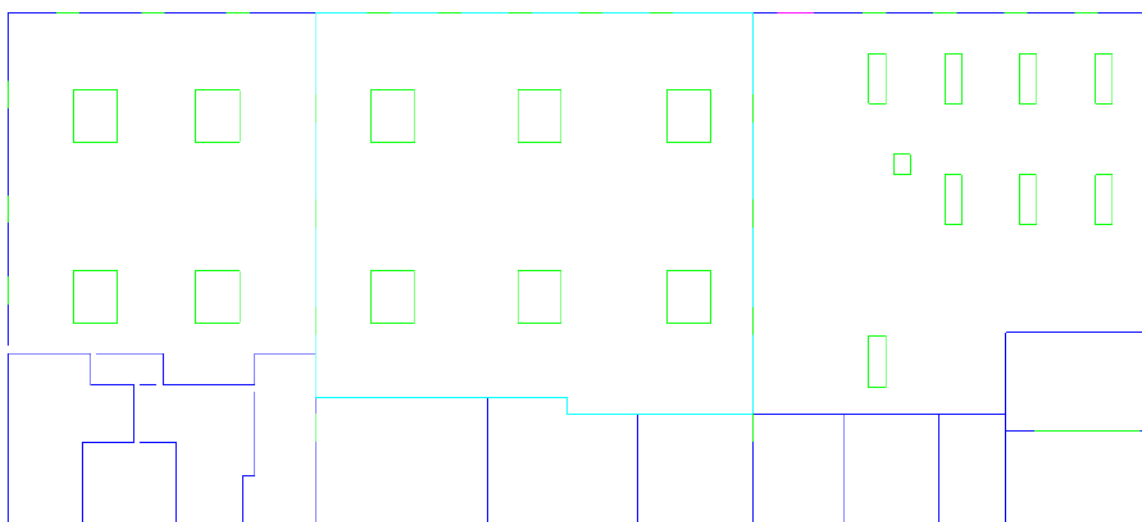


Figure 6a: Plan- Ground Floor

5. BASELINE – Target Emission Rate

The Target Emission Rate (TER) and Target Primary Energy Rate (TPER), as calculated in line with the Building Regulation 2021, represent the maximum amount of carbon a development is allowed to emit and the maximum of amount of energy it can consume. These target values include energy consumption and carbon emissions resulting from functions covered by Part L of the Building Regulations, also known as regulated emissions (space and water heating, ventilation, lighting, pumps, fans & controls).

This first run of simulation assesses the models under the notional building specifications defined under Part L to limit the fabric thermal heat losses. The assessment results in the regulated target values for energy consumption and carbon emissions i.e. the TER and TPER, by using the Government approved methodology (SBEM).

The baseline regulated CO₂ emissions for the development is presented in the table below:

Table 6: Regulated Carbon Emissions at Baseline

Model Name	BER (KgCO ₂ /m ² .annum)	BPER (KWh/m ² .annum)
Notional Building	7.42	80.12
Actual Building	16.93	181.26

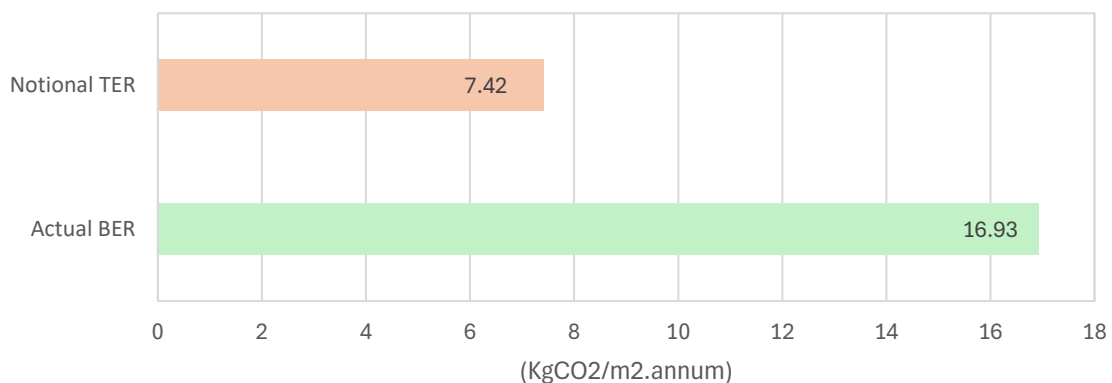


Figure 7: Actual vs. Notional BER for the proposed scheme

6. BE LEAN – Energy Efficient Design

This section outlines the project location, orientation and climatic conditions considered to determine the most feasible energy efficient measures taken in order to minimise the building's energy demand. The analysis helps reduce the energy use and CO₂ emissions beyond the Baseline benchmark on the path to achieving compliance (**Building Regulations 2021 Part L2**).

6.1 Site Location



Figure 8: 3D urban view of the site location (Source: Google Earth)

The proposed project site is located at Dorset Road in the London borough of Sutton. The local surrounding context is largely low rise commercial and residential.

6.2 Site Weather and Microclimate

The local weather microclimate usually influences buildings' energy performance. Urban design has a significant impact on microclimate and outdoor thermal comfort. Several studies in different climate regions have concluded that ventilation and shade are crucial to improve urban thermal comfort. Often the thermal conditions are improved as a consequence of good urban design including exist of proper shade and sufficient ventilation. This in turn leads to decreased occurrence of heat stress and heat-related diseases as well as grown performance of both mental and physical tasks.

London weather data has been used for thermal and energy simulations of this assessment. Figure 9 below represents the annual dry bulb temperature in London City. Maximum temperatures can reach up to 34.9°C in late Jun, and minimum temperatures can reach down to -2.1°C in late Feb. The buildings in this location are generally low rise, and the site is not expected to suffer from the Urban Heat Island effects.

The site's landscape also affects the energy demand of a building. Vegetation, landform, and any existing buildings can provide shade to a new development. For instance, if located to the south of the building, deciduous trees can be advantageous, providing shade in the summer but allowing

sunshine through in the winter when they lose their leaves. However, in this case, the context is largely urban and no dense vegetation exists in the immediate vicinity of the site.

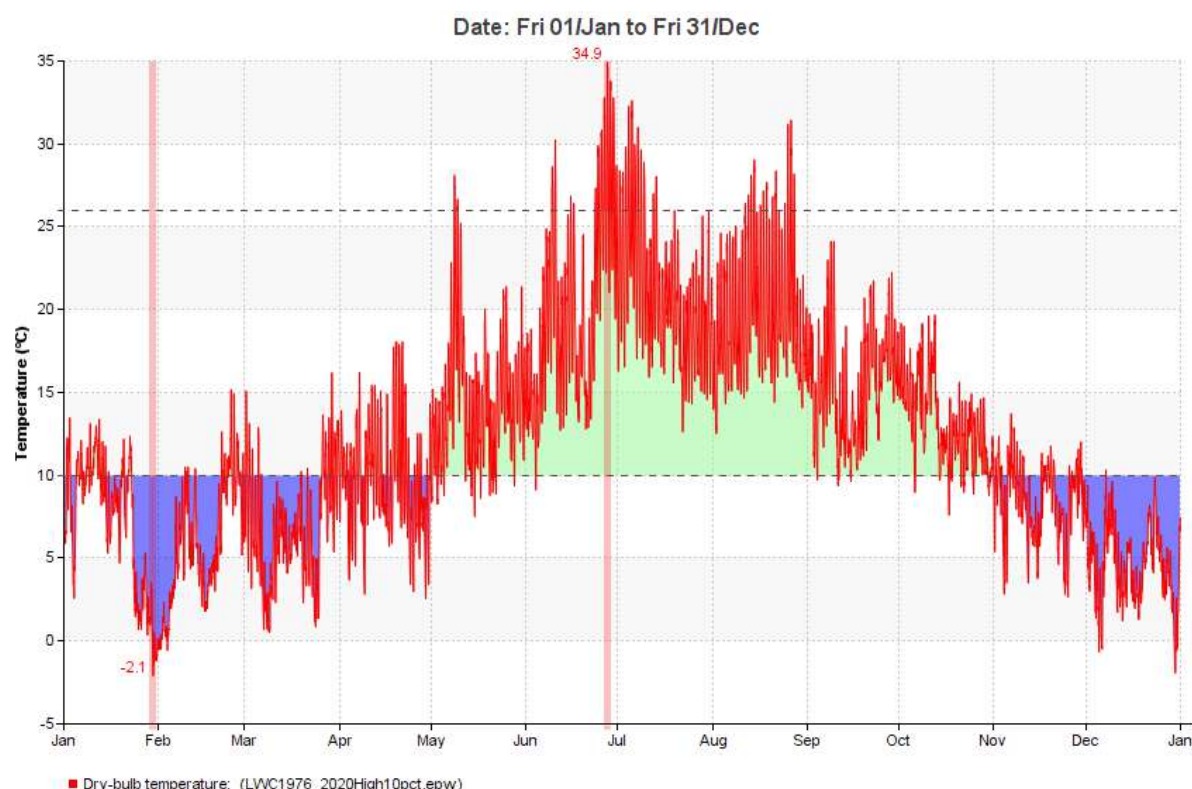


Figure 9: IES Annual Dry Bulb Temperature according to London weather file

6.3 Building Orientation, Layout and Form

Building layout, orientation and form can influence many key features of the development. The design should provide for an effective use of space and appealing layout, with opportunities to benefit from natural daylight balanced with achieving solar gain without overheating. In general, a higher thermal performance can be achieved by limiting the surface area to volume ratio as this minimises heat loss through the wall area.

It should be noted that where the building footprint is extremely tight, for example in an urban location (as is the case in this assessment), the building form and orientation may have to be dictated by the available space and not by implementation of best practice measures. Invariably, planning constraints and/or the functional relationships of specific areas will result in some measure of deep planning, thus reducing the opportunity for natural ventilation and daylight.

Despite the constraints of the tight and long site, passive solar gains and daylight into the building have been optimised. The regularly occupied spaces that have glazing include the prayer hall and multi-purpose hall. The glazing specifications assumed for the energy modelling are summarized in the table below:

Table 7: Glazing specifications

	U-value (Including frame)	G-value	VLT
Be lean	1.10	0.34	0.76

The primary facades of the proposed building face roughly East and West. The rooms are expected to receive the morning or afternoon sunlight, based on their orientation.



Figure 10: Proposed site plan.

6.4 Building Design – Energy Efficiency Measures

At the 'BE LEAN' stage of the energy hierarchy, energy efficient building elements have been incorporated into the building design. The heat loss of different building elements is dependent upon their U-value, and air tightness. Therefore, better u-values and air permeability than the minimum values set in the **Part L2 2021** have been suggested in this stage of the assessment process.

The table below shows the proposed building fabric thermal performance. It promotes a thermally efficient building fabric in order to reduce energy consumption through passive measures. This strategy optimises the building performance prior to the consideration of renewables (LZC).

Table 8: Proposed thermal performance for construction material.

Category Specification		Part L2 - 2021 specifications	Be Lean Proposed Values
U-value (W/m ² K)	Wall	0.26	0.11
	Window	1.60	1.10
	Floor	0.18	0.11
	Roof	0.18	0.11
Air Permeability (m ³ /h.m ² at 50 Pa)		8.00	2.50

In order to achieve highly efficient building envelopes such as the one being suggested above, incorporating enhanced levels of insulation within the building elements is inevitable.

6.5 Building Occupancy Type

The building occupancy type is dependent on the NCM building type. A Other: Emergency Services building type has been identified for this development. Accordingly, the occupancy capacity and profiles as well as internal gains are input in accordance with the activity type associated with each space. The NCM activity has been allocated as per the table below:

Table 9: NCM activity type selected for energy model

Space	Building area type	NCM Activity
Multipurpose Hall	D2: General assembly and Leisure (Social Club)	NCM D2Cl: Hall/lecture theatre/assembly area (SocClub)
Prayer Hall	D2: General assembly and Leisure (Social Club)	NCM D2Cl: Hall/lecture theatre/assembly area (SocClub)
Kitchen	Other: Emergency Services	NCM EmgcySvc: Food preparation area
Office	Other: Emergency Services	NCM EmgcySvc: Office
Toilet	Other: Emergency Services	NCM EmgcySvc: Toilet

6.6 Daylighting and Solar shading strategy

The glazing specification is carefully selected to ensure the internal environment is pleasant on all orientations. The spaces are recommended to make use of blinds to limit excess solar gains, particularly for the reception lobby on the ground floor as it has a large glazing area facing due South. This will help reduce the overheating risk, and subsequently the cooling demand.

6.7 Ventilation Strategy

Natural Ventilation + Heat Recovery Ventilation

All spaces will utilise trickle and boost ventilation via heat recovery units that utilise the warm

exhausted air to pre-heat the incoming supply air. This helps to significantly reduce the heat loss of the building where ventilation is the highest heat demand.

Be Lean – Active Design Measures

The table below shows the applied ‘Be Lean’ active design measures incorporated within the design to demonstrate the energy strategy approach.

Table 10: Proposed active design measures adopted at the Be Lean stage.

No.	Measure	Explanation
1	Heat recovery ventilation	The ventilation strategy utilises heat recovery ventilation as part of a mixed-mode ventilation strategy that is deployed to maximise energy efficiency. Duct insulated, Rigid duct, Specific Fan Power (SFP) 0.3 W/l/s, Heat Exchanger Efficiency 85% has been applied for this assessment.
2	On-demand control	The room by room heating and lighting controls ensure that energy efficiency is maximised in almost all spaces where the amount of air delivered is variable to suit the space.
3	Local heat network	Heating for the building is provided by a building wide heat network which features reduced flow and return temperatures, reducing thermal losses and space heat gain as compared to a traditional system. Due to the nature of the LTHW local heat network should a community heat network connection become available to the site in the future, the proposed system is compatible for connection and the connection points in place.
4	Variable speed pumping	Variable speed pumping on all secondary circuits reduces energy consumption associated with distribution significantly.
5	Lighting Efficiency	LED’s have been appraised for their suitability as part of the design development and have been found to be a cost-effective measure to reduce lighting electrical energy use.

The output from the building model confirms that upgrading the building fabric will result in the building meeting the Part L 2021 Requirements as presented in Table 12 below. The project is able to achieve 15% reduction at the Be Lean stage. This helps fulfil the criteria of $\geq 15\%$ reduction over Part L baseline from energy efficiency measures alone.

Table 11: Regulated Carbon Emissions at Be Lean stage

Model Name	BER (KgCO ₂ /m ² .annum)	BPER (KWh/m ² .annum)
Notional Building	7.42	80.12
Be Lean stage	5.73	62.20

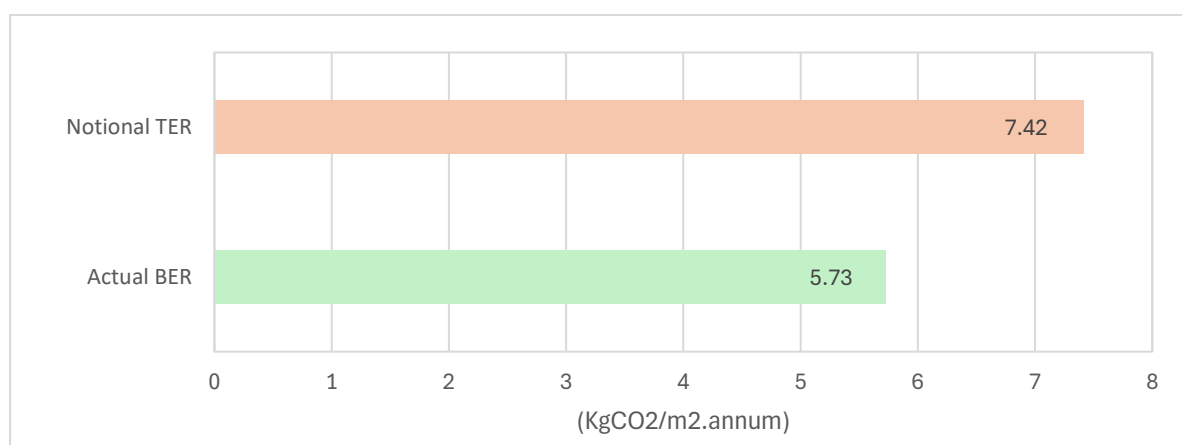


Figure 11: Notional TER vs. Be Lean BER for the proposed scheme.

7. BE CLEAN - Decentralised Energy Networks

The Energy Hierarchy encourages the use of a CHP system and connection to District Heating systems to reduce CO₂ emissions further.

7.1 Decentralised Energy Network

District and community heating systems are encouraged because they offer:

- Potential economies of scale in respect of efficiency and therefore reduced carbon emissions.
- Greater potential for future replacement with Low or Zero Carbon (LZC) technologies.

The feasibility of connecting into an existing heating network or providing the building with its own combined heat and power plant has been assessed alongside the **London Heat Map** as part of this assessment (refer figure 12 below). The study identifies that the site is not located near the existing district heating networks, with the nearest network being approximately >500 m away. This is demonstrated clearly from the London Heat Map (<http://www.londonheatmap.org.uk>).



Figure 12: Existing and potential DH Network near the site

As demonstrated by the image above, no existing Heat Networks are located close to the site. Proposed Heat Networks are planned in the area. To enable access to the nearest existing network (distance >500 m), the costs involved in extending the network would outweigh the advantages the development stands to gain. **Therefore, utilisation of a local heat network is not determined as a feasible option for this development.**

7.2 Combined Heat and Power (CHP)

The Energy Hierarchy identifies the combined heat and power (CHP) as a method of producing heat and electricity with much lower emissions than separate heat and power. Also, it encourages the

creation of district heating systems supplied by CHP. The implementation of a CHP strategy should be decided according to good practice design. Key factors for the efficient implementation of the CHP system are:

- Development with high heating load for the majority of the year.
- CHP operation based on maximum heat load for minimum 10 hours per day.
- CHP operation at maximum capacity of 90% of its operating period.

To ensure that CHP is financially viable it is essential that the unit is selected to meet the base heat load and that this load is maintained over a large proportion of the day (a figure of 14 – 17 hours per day is often quoted subject to the load profiles and gas and electricity prices) to ensure that the additional costs (maintenance) associated with running a CHP unit can be recovered. The need to run the CHP plant continuously as far as possible makes the building load profile of prime importance when reviewing the viability of such solutions and in particular the summertime heat load profile. To enable the CHP plant to run continuously when it is operating, a thermal store is often used so that excess CHP capacity can be used at a later time.

The feasibility of installing CHP has been assessed for this development. **Since this building is fall under minor development that does not require high heating loads, installing the CHP system would not be beneficial given the cost.** According to the GLA Guidance on preparing energy assessment, the development consisting around 500 residential units or more would obtain the benefits of the CHP system. **Hence the CHP system has not been considered for this development.**

Table 12: Regulated Carbon Emissions at Be Clean stage

Model Name	BER (KgCO ₂ /m ² .annum)	BPER (KWh/m ² .annum)
Notional Building	7.42	80.12
Be Clean stage	4.06	43.02

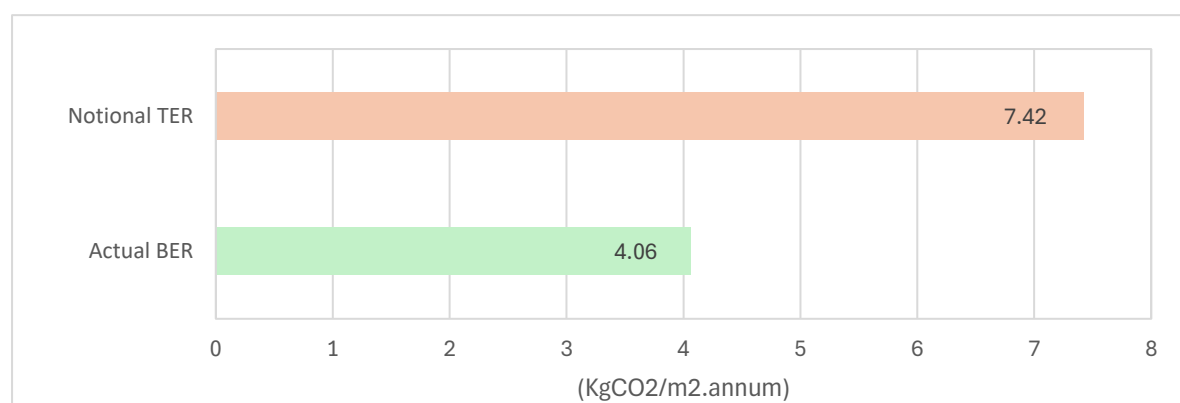


Figure 13: Notional TER vs. Be Clean BER for the proposed scheme.

8. BE GREEN – Renewable and Low-carbon Technology Energy

In this section the viable renewable energy technologies that could reduce the development's CO₂ emissions are examined. In accordance with Policy SI 2 of the London Plan, the technical feasibility and economic viability of installing each LZC technology at the development have been assessed, so as to discount any unsuitable options at an early stage.

In determining the appropriate renewable technology for the site, the following factors were considered:

- Renewable energy resources or fuel availability of the LZC technology on the site.
- Space limitations due to building design and urban location of the site.
- Capital, operating and maintenance cost.
- Planning Permission
- Implementation with regards the overall M&E design strategy for building type
- Available Grants

The table below summarises the various low zero carbon technologies considered for the projects, and we have identified that **Air Source Heat Pumps (ASHP)** and **Photovoltaic (PV)** would be the most appropriate options in this development.

Table 13: Feasibility Study of LZC Technologies

Technology Name:	Carbon Payback	Feasibility
Photovoltaic (PV)	High	HIGH
Air Source Heat Pumps (ASHP)	High	HIGH
Biomass	High	LOW
Wind Power	Low	LOW
Hydro Power	None	LOW
Solar Thermal	Low	MEDIUM
Ground Source Heat Pumps (GSHP)	Medium	LOW

8.1 Non-feasible Technology

- **Ground Source Heat Pumps (GSHP)**

A ground source heat pump would be a feasible option to meet the space heating requirements. However, it requires ground space for bore holes to extract the ground heat to be utilized for space heating requirements. In this case there is no available ground space for a borehole or trench system, the ground source loop would have to be incorporated within the foundation piles of the structure, which would result in additional cost. Hence, this option is not suitable for this development.

- **Solar Thermal**

The use of solar thermal for this development would be limited to domestic hot water only. The use of solar thermal for space heating would not be practical as it is not required when solar thermal is at its most effective during the summer months. Therefore, this system would require additional plumbing and space for hot water storage, incurring additional financial costs. Moreover, the amount of carbon offset from the system is generally lower than other technologies. Therefore, this technology is deemed to be unsuitable for this development. As only 90m² of PV would be needed to offset all regulated carbon and DHW consumption contributes less than 15%, it will not be worth to accommodate hot water storage.

- **Hydro power**

The project site is not close to natural water resources. Therefore, small scale hydro-electric will not be studied any further because of the location and the spatial limitations of the development.

- **Biomass**

A biomass system designed for this development would be fueled by wood pellets which have a high energy content. However, a biomass system would not be an appropriate technology for the site for the following reasons:

- The burning of wood pellets releases substantially more NO_x emissions when compared to similar gas boilers. As the development is situated within an urban area, the installation of a biomass boiler would further impact on the air quality in this area.
- the lack of spaces for pellet boiler and storage on the site.
- Pellets would need to be transported from local pellet suppliers, which causes carbon emissions into the air.

However, if the biomass system is considered at detailed design stage, local suppliers can be found near the site as shown in the map below (<http://biomass-suppliers-list.service.gov.uk>).

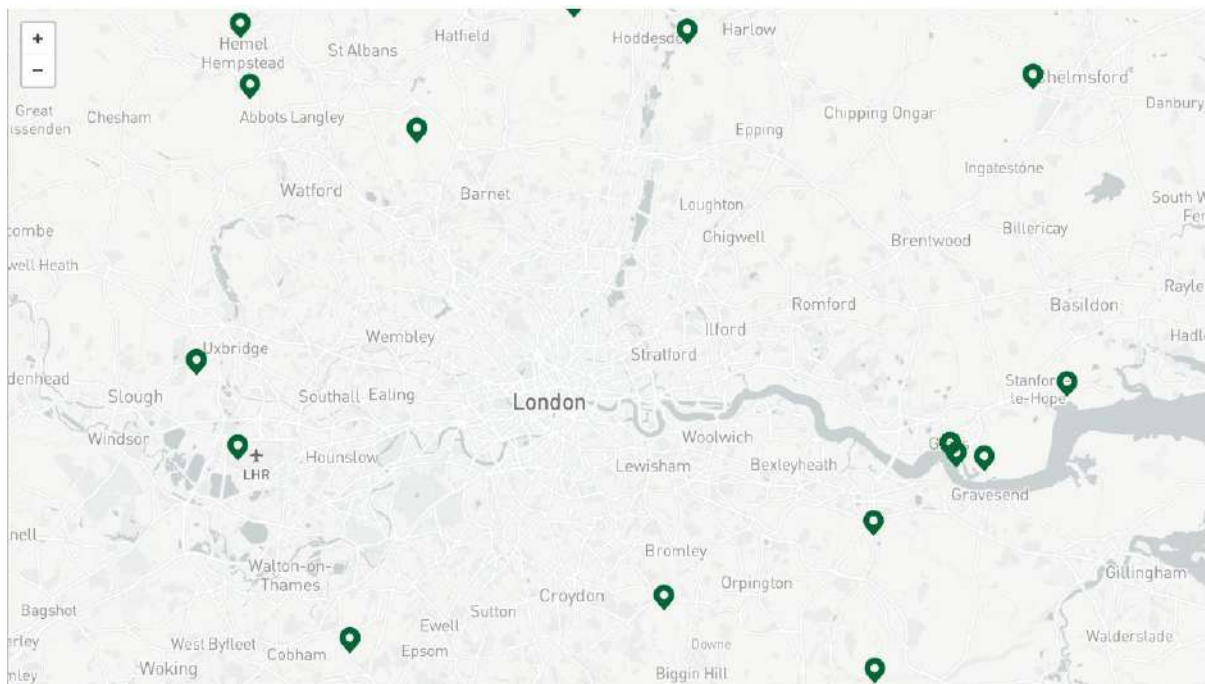


Figure 14: Biomass suppliers' locations in proximity to the project location.

• Wind Power

Wind turbines need extensive planning requirements, and they are only feasible at consistent wind speed. Moreover, since the development is located in an urban area, the site does not have sufficient wind speed to operate wind turbine at the height of 10 meters as shown below (Source: <https://www.rensmart.com/Maps#NOABL>). Hence this option has been discounted.

Location

- Latitude: 51.349800
- Longitude: -0.2018800
- Height above sea level: 9m

Wind Speeds

estimates from NOABL data

- At 10 m above ground level: 5.0 m/s
- At 25 m above ground level: 5.8 m/s
- At 45 m above ground level: 6.3 m/s

Solar

falling on optimally oriented panel

- Annual Energy per m² 1138.8 kWh
- Daily Energy per m² 3120 Wh

Please note that the NOABL windspeed dataset used here is a model of windspeeds across the country, assuming completely flat terrain. It isn't a database of measured windspeeds. Other factors such as hills, houses, trees and other obstructions in the vicinity need to be considered as well since they can have a significant effect. If the installation of a wind turbine is considered, windspeed measurements using an anemometer need to be performed to determine what the actual figures are.

8.2 Feasible Technology

• Air Source Heat Pumps (ASHP)

ASHP can meet the space heating demands on site more efficiently in comparison to gas boilers. Although this low carbon technology consumes electricity to operate, due to higher efficiency the heat output is much greater. Therefore, it has been suggested for the space heating and hot water for this project.

Another advantage to the use of ASHP technology is that it has no impact on the Air Quality credentials i.e., NO_x, Particulate Matter etc. Emissions associated with building operations, specifically those resulting from space conditioning of the building are minimized.

The design stage specifications used for energy calculations are in the table below. However, the ASHP was proposed only for simulation, detailed ASHP specifications should be provided by a mechanical engineer during the design development. The system must be certified under the Microgeneration Certification Scheme (MCS) and ideally, the Enhanced Capital Allowances criteria.

ASHP Air Distribution system recommended minimum standards.

In order to limit air leakage, ventilation ductwork should be made and assembled so as to be reasonably airtight. Ways of meeting this requirement would be to comply with the specifications given in:

- B&ES DW/144⁴³. Membership of the B&ES specialist ductwork group or the Association of Ductwork Contractors and Allied Services is one way of demonstrating suitable qualifications, or
- British Standards such as BS EN 1507:2006⁴⁴, BS EN 12237:2003⁴⁵ and BS EN 13403:2003⁴⁶.

In order to limit air leakage, air handling units should be made and assembled so as to be reasonably airtight. Ways of meeting this requirement would be to comply with Class L2 air leakage given in BS EN 1886:2007⁴⁷.

The specific fan power of air distribution systems at the design air flow rate should be no worse than in Table 35 for new and existing buildings. Specific fan power is a function of the system resistance that the fan has to overcome to provide the required flow rate. BS EN 13779 Table A8 provides guidance on system pressure drop. To minimise specific fan power it is recommended that the 'low range' is used as a design target.

Where the primary air and cooling is provided by central plant and by an air distribution system that includes the additional components listed in Table 36, the allowed specific fan powers may be increased by the amounts shown in Table 36 to account for the additional resistance.

A minimum controls package should be provided in new and existing buildings as in Table 37.

Ventilation fans driven by electric motors should comply with European Commission Regulation No 327/2011 implementing Directive 2009/125/EC with regard to ecodesign requirements for fans driven by motors with an electric input power between 125 W and 500 kW.

(Source: Non-Domestic Building Services Compliance Guide)

Table 14: ASHP system specs, as assumed for energy modelling.

Systems	General Specification	Control
Heating and Cooling	Air Source Heat Pump (SCOP 4.3, SEER 6.5)	- System SFP 1.1 W/l/s and Local supply SFP 0.3 W/l/s - Pump type – variable speed multiple pressure sensors - Heat recovery (plate heat exchanger) efficiency 0.85

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BSI Certificate Number FS 710041



Hot Water 100% demand being met by PoU • Storage losses – 5%

- **Photovoltaic (PV)**

Solar PV could be one of the suitable renewable technologies for the project due to the following reasons:

- The installation of PV is much simpler when compared to other renewable technologies.
- There is sufficient roof space available to install enough PV modules to have a significant impact on carbon emissions of the development.
- PV panels sited on the roof within an urban area are less visually intrusive when compared to wind turbines.

Table 15: Suggested solar PV specifications.

Property	General specification
Type	Monocrystalline Silicon
Orientation	South
Inclination	30°
Power output	4.63 MWh/year
Plant size	5.02 kWp
Total PV area	25 m ²

In summary, a 5.02 kWp (assuming 1000 kWh annual output per kWp) solar PV plant is recommended in order to meet the requirement of generating 10% of the building's energy needs from on-site renewables. Monocrystalline silicon panels spread over approx. 25 m² of roof space should be able to provide the required output. However, the proposed PV panels are subject to further consideration at detailed design stages. In order to qualify both the installer and the equipment, the system must be certified under the Microgeneration Certification Scheme (MCS).

Given the proposed LZC technologies on the site (ASHP and Solar PV), the overall CO₂ reduction at this stage can be calculated as shown below:

Table 15: Regulated carbon reduction at Be Green stage.

Model Name	BER (KgCO ₂ /m ² .annum)	BPER (KWh/m ² .annum)
Notional Building	7.42	80.12
Be Green stage	4.06	43.02

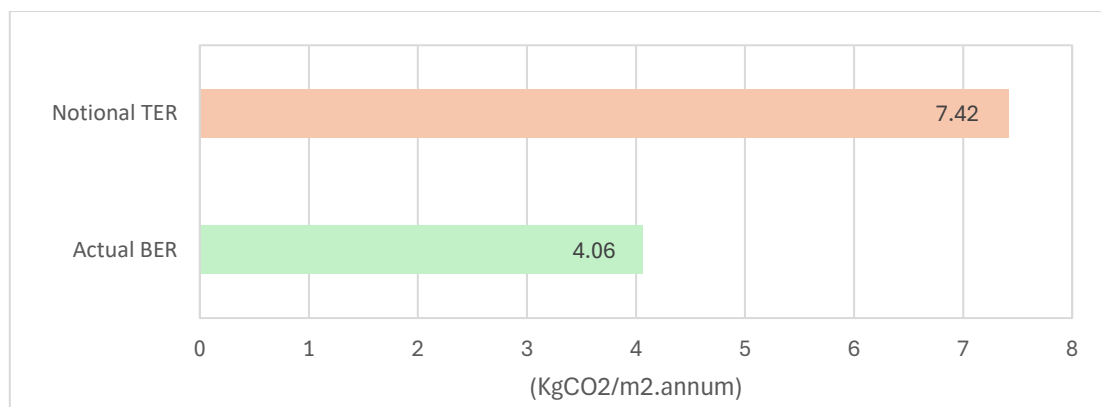


Figure 15: Notional TER vs. Be Clean BER for the proposed scheme.

9. Conclusion

This report assesses the predicted energy performance and carbon dioxide emissions of the proposed development at **Sutton Ambulance Station, Dorset Road, Sutton, London SM2 6JH**.

Based on the information provided by the design team, the study has been done for the proposed change of use from sui generis [ambulance station to community centre use with prayer facility]. The study results showed improving the building fabric thermal performance is a key to achieve building regulations compliance. Moreover, efficient lighting control units and ASHP proved to be an efficient measure for the new build to achieve the recommended **>35%** beyond **Part L2** requirements. In addition, if the inclusion of PVs is opted in the design specification, it could enhance the project sustainability status further. The carbon savings from each stage are shown in the chart below. Given the total cumulative carbon savings, the proposed development shall meet the planning requirements on the site.

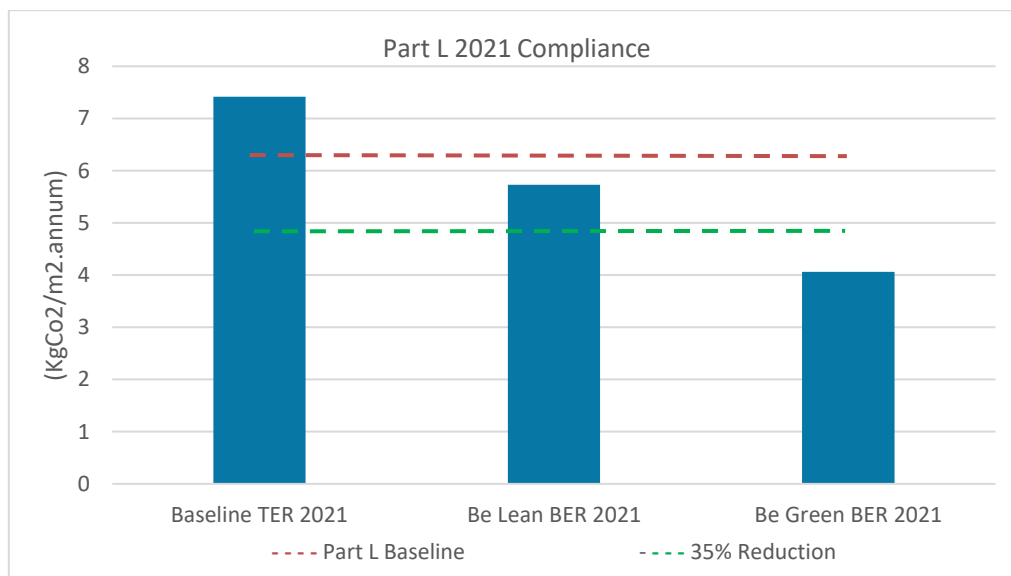


Figure 16: Carbon Emissions Reductions after each stage of the Energy Hierarchy

Table 16: Overall carbon reductions

Model Name	BER (KgCO ₂ /m ² .annum)	% Reductions from Baseline
Notional Building	7.42	
Be Lean stage	5.73	22.77%
Be Clean stage	5.73	22.77%
Be Green stage	4.06	45.28%

10. Appendix A – BRUKL reports

Project name

Former Ambulance Station _ Baseline**As designed**

Date: Wed Dec 17 09:07:53 2025

Administrative information

Building Details

Address: Sutton Ambulance Station, Dorset Road,
Sutton, London SM2 6JH

Certifier details

Name: Name**Telephone number:** Phone**Address:** Street Address, City, City

Certification tool

Calculation engine: Apache**Calculation engine version:** 7.0.30**Interface to calculation engine:** IES Virtual Environment**Interface to calculation engine version:** 7.0.30**BRUKL compliance module version:** v6.1.e.1Foundation area [m²]: 359.8The CO₂ emission and primary energy rates of the building must not exceed the targets

The building does not comply with England Building Regulations Part L 2021

Target CO ₂ emission rate (TER), kgCO ₂ /m ² annum	7.42
Building CO ₂ emission rate (BER), kgCO ₂ /m ² annum	16.93
Target primary energy rate (TPER), kWh _{PE} /m ² annum	80.12
Building primary energy rate (BPER), kWh _{PE} /m ² annum	181.26
Do the building's emission and primary energy rates exceed the targets?	BER > TER BPER > TPER

The performance of the building fabric and fixed building services should achieve reasonable overall standards of energy efficiency

Fabric element	U _a -Limit	U _a -Calc	U _i -Calc	First surface with maximum value
Walls*	0.26	0.26	0.26	SP000001:Surf[6]
Floors	0.18	0.18	0.18	SP000001:Surf[0]
Pitched roofs	0.16	-	-	No pitched roofs in building
Flat roofs	0.18	0.18	0.18	SP000001:Surf[1]
Windows** and roof windows	1.6	1.6	1.6	SP000001:Surf[2]
Rooflights***	2.2	2.2	2.2	SP000002:Surf[12]
Personnel doors^	1.6	1.6	1.6	SP000004:Surf[24]
Vehicle access & similar large doors	1.3	-	-	No vehicle access doors in building
High usage entrance doors	3	-	-	No high usage entrance doors in building
U _a -Limit = Limiting area-weighted average U-values [W/(m ² K)] U _a -Calc = Calculated area-weighted average U-values [W/(m ² K)] U _i -Calc = Calculated maximum individual element U-values [W/(m ² K)] * Automatic U-value check by the tool does not apply to curtain walls whose limiting standard is similar to that for windows. ** Display windows and similar glazing are excluded from the U-value check. *** Values for rooflights refer to the horizontal position. ^ For fire doors, limiting U-value is 1.8 W/m ² K NB: Neither roof ventilators (inc. smoke vents) nor swimming pool basins are modelled or checked against the limiting standards by the tool.				

Air permeability	Limiting standard	This building
m ³ /(h.m ²) at 50 Pa	8	8

Building services

For details on the standard values listed below, system-specific guidance, and additional regulatory requirements, refer to the Approved Documents.

Whole building lighting automatic monitoring & targeting with alarms for out-of-range values	NO
Whole building electric power factor achieved by power factor correction	<0.9

1- Heating and Cooling Via FCU Feeding ASHP

	Heating efficiency	Cooling efficiency	Radiant efficiency	SFP [W/(l/s)]	HR efficiency
This system	2.5	5	0	3	-
Standard value	2.5*	N/A	N/A	2^	N/A
Automatic monitoring & targeting with alarms for out-of-range values for this HVAC system					NO
* Standard shown is for all types >12 kW output, except absorption and gas engine heat pumps.					
^ Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.					

"No HWS in project, or hot water is provided by HVAC system"

Zone-level mechanical ventilation, exhaust, and terminal units

ID	System type in the Approved Documents
A	Local supply or extract ventilation units
B	Zonal supply system where the fan is remote from the zone
C	Zonal extract system where the fan is remote from the zone
D	Zonal balanced supply and extract ventilation system
E	Local balanced supply and extract ventilation units
F	Other local ventilation units
G	Fan assisted terminal variable air volume units
H	Fan coil units
I	Kitchen extract with the fan remote from the zone and a grease filter
NB: Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.	

Zone name	SFP [W/(l/s)]										HR efficiency	
ID of system type	A	B	C	D	E	F	G	H	I			
Standard value	0.3	1.1	0.5	2.3	2	0.5	0.5	0.4	1		Zone	Standard
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Mens Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Multi Purpose Hall	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Office	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Mens Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Ladies Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Ladies Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	-	N/A

General lighting and display lighting		General luminaire		Display light source	
Zone name		Efficacy [lm/W]		Efficacy [lm/W]	Power density [W/m²]
	Standard value	95		80	0.3
L00: Kitchen		95		-	-
L00: Mens Prayer Hall		95		-	-
L00: Multi Purpouse Hall		95		-	-
L00: Office		95		-	-
L00: Circulation Area		95		-	-
L00: Disabled Toilet		95		-	-
L00: Storage		95		-	-
L00: Circulation Area		95		-	-
L00: Mens Ablution/ Toilet		95		-	-
L00: Ladies Prayer Hall		95		-	-
L00: Kitchen		95		-	-
L00: Storage		95		-	-
L00: Ladies Ablution/ Toilet		95		-	-
L00: Circulation Area		95		-	-
L00: Disabled Toilet		95		-	-

The spaces in the building should have appropriate passive control measures to limit solar gains in summer

Zone	Solar gain limit exceeded? (%)	Internal blinds used?
L00: Kitchen	NO (-3.1%)	NO
L00: Mens Prayer Hall	NO (-29.7%)	NO
L00: Multi Purpouse Hall	NO (-35.8%)	NO
L00: Office	NO (-58.6%)	NO
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A
L00: Storage	N/A	N/A
L00: Circulation Area	NO (-74.8%)	NO
L00: Mens Ablution/ Toilet	NO (-77.9%)	NO
L00: Ladies Prayer Hall	NO (-34.9%)	NO
L00: Kitchen	NO (-48.2%)	NO
L00: Storage	N/A	N/A
L00: Ladies Ablution/ Toilet	N/A	N/A
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A

Regulation 25A: Consideration of high efficiency alternative energy systems

Were alternative energy systems considered and analysed as part of the design process?	YES
Is evidence of such assessment available as a separate submission?	YES
Are any such measures included in the proposed design?	YES

Technical Data Sheet (Actual vs. Notional Building)

Building Global Parameters

	Actual	Notional
Floor area [m ²]	359.8	359.8
External area [m ²]	1026	1026
Weather	LON	LON
Infiltration [m ³ /hm ² @ 50Pa]	8	3
Average conductance [W/K]	345.54	347.35
Average U-value [W/m ² K]	0.34	0.34
Alpha value* [%]	25	10

* Percentage of the building's average heat transfer coefficient which is due to thermal bridging

Building Use

% Area Building Type

Retail/Financial and Professional Services
 Restaurants and Cafes/Drinking Establishments/Takeaways
 Offices and Workshop Businesses
 General Industrial and Special Industrial Groups
 Storage or Distribution
 Hotels
 Residential Institutions: Hospitals and Care Homes
 Residential Institutions: Residential Schools
 Residential Institutions: Universities and Colleges
 Secure Residential Institutions
 Residential Spaces
 Non-residential Institutions: Community/Day Centre
 Non-residential Institutions: Libraries, Museums, and Galleries
 Non-residential Institutions: Education
 Non-residential Institutions: Primary Health Care Building
 Non-residential Institutions: Crown and County Courts
77 General Assembly and Leisure, Night Clubs, and Theatres
 Others: Passenger Terminals
23 Others: Emergency Services
 Others: Miscellaneous 24hr Activities
 Others: Car Parks 24 hrs
 Others: Stand Alone Utility Block

Energy Consumption by End Use [kWh/m²]

	Actual	Notional
Heating	37.16	9.22
Cooling	6.03	4
Auxiliary	44.26	20.15
Lighting	25.04	15.12
Hot water	6.16	5.56
Equipment*	15.1	15.1
TOTAL **	118.64	54.05

* Energy used by equipment does not count towards the total for consumption or calculating emissions.

** Total is net of any electrical energy displaced by CHP generators, if applicable.

Energy Production by Technology [kWh/m²]

	Actual	Notional
Photovoltaic systems	0	0
Wind turbines	0	0
CHP generators	0	0
Solar thermal systems	0	0
<i>Displaced electricity</i>	<i>0</i>	<i>0</i>

Energy & CO₂ Emissions Summary

	Actual	Notional
Heating + cooling demand [MJ/m ²]	382.83	158.93
Primary energy [kWh _{PE} /m ²]	181.26	80.11
Total emissions [kg/m ²]	16.93	7.42

HVAC Systems Performance										
System Type	Heat dem MJ/m2	Cool dem MJ/m2	Heat con kWh/m2	Cool con kWh/m2	Aux con kWh/m2	Heat SSEEF	Cool SSEER	Heat gen SEFF	Cool gen SEER	
[ST] Fan coil systems, [HS] ASHP, [HFT] Electricity, [CFT] Electricity										
	Actual	310	72.8	37.2	6	44.3	2.32	3.36	2.5	5
	Notional	92.2	66.7	9.2	4	20.2	2.78	4.63	----	----

Key to terms

- Heat dem [MJ/m2] = Heating energy demand
- Cool dem [MJ/m2] = Cooling energy demand
- Heat con [kWh/m2] = Heating energy consumption
- Cool con [kWh/m2] = Cooling energy consumption
- Aux con [kWh/m2] = Auxiliary energy consumption
- Heat SSEFF = Heating system seasonal efficiency (for notional building, value depends on activity glazing class)
- Cool SSEER = Cooling system seasonal energy efficiency ratio
- Heat gen SSEFF = Heating generator seasonal efficiency
- Cool gen SSEER = Cooling generator seasonal energy efficiency ratio
- ST = System type
- HS = Heat source
- HFT = Heating fuel type
- CFT = Cooling fuel type

BRUKL Output Document



Compliance with England Building Regulations Part L 2021

Project name

Former Ambulance Station _ Be Green

As designed

Date: Wed Dec 17 09:17:33 2025

Administrative information

Building Details

Address: Sutton Ambulance Station, Dorset Road,
Sutton, London SM2 6JH

Certifier details

Name: Name

Telephone number: Phone

Address: Street Address, City, Postcode

Certification tool

Calculation engine: Apache

Calculation engine version: 7.0.30

Interface to calculation engine: IES Virtual Environment

Interface to calculation engine version: 7.0.30

BRUKL compliance module version: v6.1.e.1

Foundation area [m²]: 359.8

The CO₂ emission and primary energy rates of the building must not exceed the targets

Target CO ₂ emission rate (TER), kgCO ₂ /m ² annum	6.9
Building CO ₂ emission rate (BER), kgCO ₂ /m ² annum	4.06
Target primary energy rate (TPER), kWh _{PE} /m ² annum	74.44
Building primary energy rate (BPER), kWh _{PE} /m ² annum	43.2
Do the building's emission and primary energy rates exceed the targets?	BER ≤ TER BPER ≤ TPER

The performance of the building fabric and fixed building services should achieve reasonable overall standards of energy efficiency

Fabric element	U _a -Limit	U _a -Calc	U _i -Calc	First surface with maximum value
Walls*	0.26	0.11	0.11	SP000001:Surf[6]
Floors	0.18	0.11	0.11	SP000001:Surf[0]
Pitched roofs	0.16	-	-	No pitched roofs in building
Flat roofs	0.18	0.11	0.11	SP000001:Surf[1]
Windows** and roof windows	1.6	1.1	1.1	SP000001:Surf[2]
Rooflights***	2.2	1.1	1.1	SP000002:Surf[12]
Personnel doors^	1.6	1.1	1.1	SP000004:Surf[24]
Vehicle access & similar large doors	1.3	-	-	No vehicle access doors in building
High usage entrance doors	3	-	-	No high usage entrance doors in building

U_a-Limit = Limiting area-weighted average U-values [W/(m²K)]

U_i-Calc = Calculated maximum individual element U-values [W/(m²K)]

U_a-Calc = Calculated area-weighted average U-values [W/(m²K)]

* Automatic U-value check by the tool does not apply to curtain walls whose limiting standard is similar to that for windows.

** Display windows and similar glazing are excluded from the U-value check.

*** Values for rooflights refer to the horizontal position.

^ For fire doors, limiting U-value is 1.8 W/m²K

NB: Neither roof ventilators (inc. smoke vents) nor swimming pool basins are modelled or checked against the limiting standards by the tool.

Air permeability	Limiting standard	This building
m ³ /(h.m ²) at 50 Pa	8	3

Building services

For details on the standard values listed below, system-specific guidance, and additional regulatory requirements, refer to the Approved Documents.

Whole building lighting automatic monitoring & targeting with alarms for out-of-range values	YES
Whole building electric power factor achieved by power factor correction	>0.95

1- Heating and Cooling Via FCU Feeding ASHP

	Heating efficiency	Cooling efficiency	Radiant efficiency	SFP [W/(l/s)]	HR efficiency
This system	2.5	5	0	1.1	0.85
Standard value	2.5*	N/A	N/A	2^	N/A
Automatic monitoring & targeting with alarms for out-of-range values for this HVAC system					YES
* Standard shown is for all types >12 kW output, except absorption and gas engine heat pumps.					
^ Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.					

"No HWS in project, or hot water is provided by HVAC system"

Zone-level mechanical ventilation, exhaust, and terminal units

ID	System type in the Approved Documents
A	Local supply or extract ventilation units
B	Zonal supply system where the fan is remote from the zone
C	Zonal extract system where the fan is remote from the zone
D	Zonal balanced supply and extract ventilation system
E	Local balanced supply and extract ventilation units
F	Other local ventilation units
G	Fan assisted terminal variable air volume units
H	Fan coil units
I	Kitchen extract with the fan remote from the zone and a grease filter
NB: Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.	

Zone name	SFP [W/(l/s)]									HR efficiency	
ID of system type	A	B	C	D	E	F	G	H	I	Zone	Standard
Standard value	0.3	1.1	0.5	2.3	2	0.5	0.5	0.4	1		
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Multi Purpose Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Office	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A

General lighting and display lighting		General luminaire		Display light source	
Zone name		Efficacy [lm/W]		Efficacy [lm/W]	Power density [W/m ²]
	Standard value	95		80	0.3
L00: Kitchen		150		-	-
L00: Mens Prayer Hall		150		-	-
L00: Multi Purpouse Hall		150		-	-
L00: Office		150		-	-
L00: Circulation Area		150		-	-
L00: Disabled Toilet		150		-	-
L00: Storage		150		-	-
L00: Circulation Area		150		-	-
L00: Mens Ablution/ Toilet		150		-	-
L00: Ladies Prayer Hall		150		-	-
L00: Kitchen		150		-	-
L00: Storage		150		-	-
L00: Ladies Ablution/ Toilet		150		-	-
L00: Circulation Area		150		-	-
L00: Disabled Toilet		150		-	-

The spaces in the building should have appropriate passive control measures to limit solar gains in summer

Zone	Solar gain limit exceeded? (%)	Internal blinds used?
L00: Kitchen	NO (-32.2%)	NO
L00: Mens Prayer Hall	NO (-46.2%)	NO
L00: Multi Purpouse Hall	NO (-49.6%)	NO
L00: Office	NO (-71.8%)	NO
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A
L00: Storage	N/A	N/A
L00: Circulation Area	NO (-82.9%)	NO
L00: Mens Ablution/ Toilet	NO (-85%)	NO
L00: Ladies Prayer Hall	NO (-45.8%)	NO
L00: Kitchen	NO (-64.7%)	NO
L00: Storage	N/A	N/A
L00: Ladies Ablution/ Toilet	N/A	N/A
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A

Regulation 25A: Consideration of high efficiency alternative energy systems

Were alternative energy systems considered and analysed as part of the design process?	YES
Is evidence of such assessment available as a separate submission?	YES
Are any such measures included in the proposed design?	YES

Technical Data Sheet (Actual vs. Notional Building)

Building Global Parameters

	Actual	Notional
Floor area [m ²]	359.8	359.8
External area [m ²]	1026	1026
Weather	LON	LON
Infiltration [m ³ /hm ² @ 50Pa]	3	3
Average conductance [W/K]	201.27	347.35
Average U-value [W/m ² K]	0.2	0.34
Alpha value* [%]	12.7	10

* Percentage of the building's average heat transfer coefficient which is due to thermal bridging

Building Use

% Area Building Type

Retail/Financial and Professional Services
 Restaurants and Cafes/Drinking Establishments/Takeaways
 Offices and Workshop Businesses
 General Industrial and Special Industrial Groups
 Storage or Distribution
 Hotels
 Residential Institutions: Hospitals and Care Homes
 Residential Institutions: Residential Schools
 Residential Institutions: Universities and Colleges
 Secure Residential Institutions
 Residential Spaces
 Non-residential Institutions: Community/Day Centre
 Non-residential Institutions: Libraries, Museums, and Galleries
 Non-residential Institutions: Education
 Non-residential Institutions: Primary Health Care Building
 Non-residential Institutions: Crown and County Courts

77 General Assembly and Leisure, Night Clubs, and Theatres

Others: Passenger Terminals

23 Others: Emergency Services

Others: Miscellaneous 24hr Activities

Others: Car Parks 24 hrs

Others: Stand Alone Utility Block

Energy Consumption by End Use [kWh/m²]

	Actual	Notional
Heating	6.68	9.99
Cooling	4.32	3.79
Auxiliary	19.78	19.96
Lighting	5.45	10.89
Hot water	5.85	5.56
Equipment*	15.1	15.1
TOTAL **	42.08	50.19

* Energy used by equipment does not count towards the total for consumption or calculating emissions.

** Total is net of any electrical energy displaced by CHP generators, if applicable.

Energy Production by Technology [kWh/m²]

	Actual	Notional
Photovoltaic systems	12.88	0
Wind turbines	0	0
CHP generators	0	0
Solar thermal systems	0	0
<i>Displaced electricity</i>	<i>12.88</i>	<i>0</i>

Energy & CO₂ Emissions Summary

	Actual	Notional
Heating + cooling demand [MJ/m ²]	118.46	163.11
Primary energy [kWh _{PE} /m ²]	43.2	74.44
Total emissions [kg/m ²]	4.06	6.9

HVAC Systems Performance										
System Type	Heat dem MJ/m2	Cool dem MJ/m2	Heat con kWh/m2	Cool con kWh/m2	Aux con kWh/m2	Heat SSEFF	Cool SSEER	Heat gen SEFF	Cool gen SEER	
[ST] Fan coil systems, [HS] ASHP, [HFT] Electricity, [CFT] Electricity										
	Actual	53.9	64.5	6.7	4.3	19.8	2.24	4.15	2.5	5
	Notional	100	63.2	10	3.8	20	2.78	4.63	----	----

Key to terms	
Heat dem [MJ/m2]	= Heating energy demand
Cool dem [MJ/m2]	= Cooling energy demand
Heat con [kWh/m2]	= Heating energy consumption
Cool con [kWh/m2]	= Cooling energy consumption
Aux con [kWh/m2]	= Auxiliary energy consumption
Heat SSEFF	= Heating system seasonal efficiency (for notional building, value depends on activity glazing class)
Cool SSEER	= Cooling system seasonal energy efficiency ratio
Heat gen SSEFF	= Heating generator seasonal efficiency
Cool gen SSEER	= Cooling generator seasonal energy efficiency ratio
ST	= System type
HS	= Heat source
HFT	= Heating fuel type
CFT	= Cooling fuel type

Project name

Former Ambulance Station**As designed****Date:** Mon Dec 1 19:18:57 2025**Administrative information****Building Details**

Address: Sutton Ambulance Station, Dorset Road,
Sutton, London SM2 6JH

Certifier details**Name:** Name**Telephone number:** Phone**Address:** Street Address, City, Postcode**Certification tool****Calculation engine:** Apache**Calculation engine version:** 7.0.30**Interface to calculation engine:** IES Virtual Environment**Interface to calculation engine version:** 7.0.30**BRUKL compliance module version:** v6.1.e.1**Foundation area [m²]:** 359.8**The CO₂ emission and primary energy rates of the building must not exceed the targets**

Target CO ₂ emission rate (TER), kgCO ₂ /m ² annum	6.9
Building CO ₂ emission rate (BER), kgCO ₂ /m ² annum	5.73
Target primary energy rate (TPER), kWh _{PE} /m ² annum	74.44
Building primary energy rate (BPER), kWh _{PE} /m ² annum	62.2
Do the building's emission and primary energy rates exceed the targets?	BER ≤ TER BPER ≤ TPER

The performance of the building fabric and fixed building services should achieve reasonable overall standards of energy efficiency

Fabric element	U _{a-Limit}	U _{a-Calc}	U _{i-Calc}	First surface with maximum value
Walls*	0.26	0.11	0.11	SP000001:Surf[6]
Floors	0.18	0.11	0.11	SP000001:Surf[0]
Pitched roofs	0.16	-	-	No pitched roofs in building
Flat roofs	0.18	0.11	0.11	SP000001:Surf[1]
Windows** and roof windows	1.6	1.1	1.1	SP000001:Surf[2]
Rooflights***	2.2	1.1	1.1	SP000002:Surf[12]
Personnel doors^	1.6	1.1	1.1	SP000004:Surf[24]
Vehicle access & similar large doors	1.3	-	-	No vehicle access doors in building
High usage entrance doors	3	-	-	No high usage entrance doors in building

U_{a-Limit} = Limiting area-weighted average U-values [W/(m²K)]U_{i-Calc} = Calculated maximum individual element U-values [W/(m²K)]U_{a-Calc} = Calculated area-weighted average U-values [W/(m²K)]

* Automatic U-value check by the tool does not apply to curtain walls whose limiting standard is similar to that for windows.

** Display windows and similar glazing are excluded from the U-value check.

*** Values for rooflights refer to the horizontal position.

^ For fire doors, limiting U-value is 1.8 W/m²K

NB: Neither roof ventilators (inc. smoke vents) nor swimming pool basins are modelled or checked against the limiting standards by the tool.

Air permeability	Limiting standard	This building
m ³ /(h.m ²) at 50 Pa	8	3

Building services

For details on the standard values listed below, system-specific guidance, and additional regulatory requirements, refer to the Approved Documents.

Whole building lighting automatic monitoring & targeting with alarms for out-of-range values	YES
Whole building electric power factor achieved by power factor correction	>0.95

1- Heating and Cooling Via FCU Feeding ASHP

	Heating efficiency	Cooling efficiency	Radiant efficiency	SFP [W/(l/s)]	HR efficiency
This system	2.5	5	0	1.1	0.85
Standard value	2.5*	N/A	N/A	2^	N/A
Automatic monitoring & targeting with alarms for out-of-range values for this HVAC system					YES
* Standard shown is for all types >12 kW output, except absorption and gas engine heat pumps.					
^ Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.					

"No HWS in project, or hot water is provided by HVAC system"

Zone-level mechanical ventilation, exhaust, and terminal units

ID	System type in the Approved Documents
A	Local supply or extract ventilation units
B	Zonal supply system where the fan is remote from the zone
C	Zonal extract system where the fan is remote from the zone
D	Zonal balanced supply and extract ventilation system
E	Local balanced supply and extract ventilation units
F	Other local ventilation units
G	Fan assisted terminal variable air volume units
H	Fan coil units
I	Kitchen extract with the fan remote from the zone and a grease filter
NB: Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.	

Zone name	SFP [W/(l/s)]									HR efficiency	
ID of system type	A	B	C	D	E	F	G	H	I	Zone	Standard
Standard value	0.3	1.1	0.5	2.3	2	0.5	0.5	0.4	1		
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Multi Purpose Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Office	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A

General lighting and display lighting		General luminaire	Display light source	
Zone name		Efficacy [lm/W]	Efficacy [lm/W]	Power density [W/m ²]
	Standard value	95	80	0.3
L00: Kitchen		150	-	-
L00: Mens Prayer Hall		150	-	-
L00: Multi Purpose Hall		150	-	-
L00: Office		150	-	-
L00: Circulation Area		150	-	-
L00: Disabled Toilet		150	-	-
L00: Storage		150	-	-
L00: Circulation Area		150	-	-
L00: Mens Ablution/ Toilet		150	-	-
L00: Ladies Prayer Hall		150	-	-
L00: Kitchen		150	-	-
L00: Storage		150	-	-
L00: Ladies Ablution/ Toilet		150	-	-
L00: Circulation Area		150	-	-
L00: Disabled Toilet		150	-	-

The spaces in the building should have appropriate passive control measures to limit solar gains in summer

Zone	Solar gain limit exceeded? (%)	Internal blinds used?
L00: Kitchen	NO (-32.2%)	NO
L00: Mens Prayer Hall	NO (-46.2%)	NO
L00: Multi Purpose Hall	NO (-49.6%)	NO
L00: Office	NO (-71.8%)	NO
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A
L00: Storage	N/A	N/A
L00: Circulation Area	NO (-82.9%)	NO
L00: Mens Ablution/ Toilet	NO (-85%)	NO
L00: Ladies Prayer Hall	NO (-45.8%)	NO
L00: Kitchen	NO (-64.7%)	NO
L00: Storage	N/A	N/A
L00: Ladies Ablution/ Toilet	N/A	N/A
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A

Regulation 25A: Consideration of high efficiency alternative energy systems

Were alternative energy systems considered and analysed as part of the design process?	YES
Is evidence of such assessment available as a separate submission?	YES
Are any such measures included in the proposed design?	YES

Technical Data Sheet (Actual vs. Notional Building)

Building Global Parameters

	Actual	Notional
Floor area [m ²]	359.8	359.8
External area [m ²]	1026	1026
Weather	LON	LON
Infiltration [m ³ /hm ² @ 50Pa]	3	3
Average conductance [W/K]	201.27	347.35
Average U-value [W/m ² K]	0.2	0.34
Alpha value* [%]	12.7	10

* Percentage of the building's average heat transfer coefficient which is due to thermal bridging

Building Use

% Area Building Type

Retail/Financial and Professional Services
 Restaurants and Cafes/Drinking Establishments/Takeaways
 Offices and Workshop Businesses
 General Industrial and Special Industrial Groups
 Storage or Distribution
 Hotels
 Residential Institutions: Hospitals and Care Homes
 Residential Institutions: Residential Schools
 Residential Institutions: Universities and Colleges
 Secure Residential Institutions
 Residential Spaces
 Non-residential Institutions: Community/Day Centre
 Non-residential Institutions: Libraries, Museums, and Galleries
 Non-residential Institutions: Education
 Non-residential Institutions: Primary Health Care Building
 Non-residential Institutions: Crown and County Courts

77 General Assembly and Leisure, Night Clubs, and Theatres

Others: Passenger Terminals

23 Others: Emergency Services

Others: Miscellaneous 24hr Activities

Others: Car Parks 24 hrs

Others: Stand Alone Utility Block

Energy Consumption by End Use [kWh/m²]

	Actual	Notional
Heating	6.68	9.99
Cooling	4.32	3.79
Auxiliary	19.78	19.96
Lighting	5.45	10.89
Hot water	5.85	5.56
Equipment*	15.1	15.1
TOTAL **	42.08	50.19

* Energy used by equipment does not count towards the total for consumption or calculating emissions.

** Total is net of any electrical energy displaced by CHP generators, if applicable.

Energy Production by Technology [kWh/m²]

	Actual	Notional
Photovoltaic systems	0	0
Wind turbines	0	0
CHP generators	0	0
Solar thermal systems	0	0
<i>Displaced electricity</i>	<i>0</i>	<i>0</i>

Energy & CO₂ Emissions Summary

	Actual	Notional
Heating + cooling demand [MJ/m ²]	118.46	163.11
Primary energy [kWh _{PE} /m ²]	62.2	74.44
Total emissions [kg/m ²]	5.73	6.9

HVAC Systems Performance										
System Type	Heat dem MJ/m2	Cool dem MJ/m2	Heat con kWh/m2	Cool con kWh/m2	Aux con kWh/m2	Heat SSEFF	Cool SSEER	Heat gen SEFF	Cool gen SEER	
[ST] Fan coil systems, [HS] ASHP, [HFT] Electricity, [CFT] Electricity										
	Actual	53.9	64.5	6.7	4.3	19.8	2.24	4.15	2.5	5
	Notional	100	63.2	10	3.8	20	2.78	4.63	----	----

Key to terms	
Heat dem [MJ/m2]	= Heating energy demand
Cool dem [MJ/m2]	= Cooling energy demand
Heat con [kWh/m2]	= Heating energy consumption
Cool con [kWh/m2]	= Cooling energy consumption
Aux con [kWh/m2]	= Auxiliary energy consumption
Heat SSEFF	= Heating system seasonal efficiency (for notional building, value depends on activity glazing class)
Cool SSEER	= Cooling system seasonal energy efficiency ratio
Heat gen SSEFF	= Heating generator seasonal efficiency
Cool gen SSEER	= Cooling generator seasonal energy efficiency ratio
ST	= System type
HS	= Heat source
HFT	= Heating fuel type
CFT	= Cooling fuel type

Project name

Former Ambulance Station _ NZC

As designed

Date: Wed Dec 17 09:47:53 2025

Administrative information

Building Details

Address: Sutton Ambulance Station, Dorset Road,
Sutton, London SM2 6JH

Certifier details

Name: Name
Telephone number: Phone
Address: Street Address, City, Postcode

Certification tool

Calculation engine: Apache
Calculation engine version: 7.0.30
Interface to calculation engine: IES Virtual Environment
Interface to calculation engine version: 7.0.30
BRUKL compliance module version: v6.1.e.1

Foundation area [m²]: 359.8The CO₂ emission and primary energy rates of the building must not exceed the targets

Target CO ₂ emission rate (TER), kgCO ₂ /m ² annum	6.9
Building CO ₂ emission rate (BER), kgCO ₂ /m ² annum	-0.28
Target primary energy rate (TPER), kWh _{PE} /m ² annum	74.44
Building primary energy rate (BPER), kWh _{PE} /m ² annum	-6.21
Do the building's emission and primary energy rates exceed the targets?	BER =< TER BPER =< TPER

The performance of the building fabric and fixed building services should achieve reasonable overall standards of energy efficiency

Fabric element	U _a -Limit	U _a -Calc	U _i -Calc	First surface with maximum value
Walls*	0.26	0.11	0.11	SP000001:Surf[6]
Floors	0.18	0.11	0.11	SP000001:Surf[0]
Pitched roofs	0.16	-	-	No pitched roofs in building
Flat roofs	0.18	0.11	0.11	SP000001:Surf[1]
Windows** and roof windows	1.6	1.1	1.1	SP000001:Surf[2]
Rooflights***	2.2	1.1	1.1	SP000002:Surf[12]
Personnel doors^	1.6	1.1	1.1	SP000004:Surf[24]
Vehicle access & similar large doors	1.3	-	-	No vehicle access doors in building
High usage entrance doors	3	-	-	No high usage entrance doors in building

U_a-Limit = Limiting area-weighted average U-values [W/(m²K)]U_i-Calc = Calculated maximum individual element U-values [W/(m²K)]U_a-Calc = Calculated area-weighted average U-values [W/(m²K)]

* Automatic U-value check by the tool does not apply to curtain walls whose limiting standard is similar to that for windows.

** Display windows and similar glazing are excluded from the U-value check.

*** Values for rooflights refer to the horizontal position.

^ For fire doors, limiting U-value is 1.8 W/m²K

NB: Neither roof ventilators (inc. smoke vents) nor swimming pool basins are modelled or checked against the limiting standards by the tool.

Air permeability	Limiting standard	This building
m ³ /(h.m ²) at 50 Pa	8	3

Building services

For details on the standard values listed below, system-specific guidance, and additional regulatory requirements, refer to the Approved Documents.

Whole building lighting automatic monitoring & targeting with alarms for out-of-range values	YES
Whole building electric power factor achieved by power factor correction	>0.95

1- Heating and Cooling Via FCU Feeding ASHP

	Heating efficiency	Cooling efficiency	Radiant efficiency	SFP [W/(l/s)]	HR efficiency
This system	2.5	5	0	1.1	0.85
Standard value	2.5*	N/A	N/A	2^	N/A
Automatic monitoring & targeting with alarms for out-of-range values for this HVAC system					YES
* Standard shown is for all types >12 kW output, except absorption and gas engine heat pumps.					
^ Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.					

"No HWS in project, or hot water is provided by HVAC system"

Zone-level mechanical ventilation, exhaust, and terminal units

ID	System type in the Approved Documents
A	Local supply or extract ventilation units
B	Zonal supply system where the fan is remote from the zone
C	Zonal extract system where the fan is remote from the zone
D	Zonal balanced supply and extract ventilation system
E	Local balanced supply and extract ventilation units
F	Other local ventilation units
G	Fan assisted terminal variable air volume units
H	Fan coil units
I	Kitchen extract with the fan remote from the zone and a grease filter
NB: Limiting SFP may be increased by the amounts specified in the Approved Documents if the installation includes particular components.	

Zone name	SFP [W/(l/s)]									HR efficiency	
ID of system type	A	B	C	D	E	F	G	H	I	Zone	Standard
Standard value	0.3	1.1	0.5	2.3	2	0.5	0.5	0.4	1		
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Multi Purpose Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Office	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Mens Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Prayer Hall	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Kitchen	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Storage	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Ladies Ablution/ Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Circulation Area	-	-	-	-	-	-	-	0.3	-	-	N/A
L00: Disabled Toilet	-	-	-	-	-	-	-	0.3	-	-	N/A

General lighting and display lighting		General luminaire		Display light source	
Zone name		Efficacy [lm/W]		Efficacy [lm/W]	Power density [W/m ²]
	Standard value	95		80	0.3
L00: Kitchen		150		-	-
L00: Mens Prayer Hall		150		-	-
L00: Multi Purpouse Hall		150		-	-
L00: Office		150		-	-
L00: Circulation Area		150		-	-
L00: Disabled Toilet		150		-	-
L00: Storage		150		-	-
L00: Circulation Area		150		-	-
L00: Mens Ablution/ Toilet		150		-	-
L00: Ladies Prayer Hall		150		-	-
L00: Kitchen		150		-	-
L00: Storage		150		-	-
L00: Ladies Ablution/ Toilet		150		-	-
L00: Circulation Area		150		-	-
L00: Disabled Toilet		150		-	-

The spaces in the building should have appropriate passive control measures to limit solar gains in summer

Zone	Solar gain limit exceeded? (%)	Internal blinds used?
L00: Kitchen	NO (-32.2%)	NO
L00: Mens Prayer Hall	NO (-46.2%)	NO
L00: Multi Purpouse Hall	NO (-49.6%)	NO
L00: Office	NO (-71.8%)	NO
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A
L00: Storage	N/A	N/A
L00: Circulation Area	NO (-82.9%)	NO
L00: Mens Ablution/ Toilet	NO (-85%)	NO
L00: Ladies Prayer Hall	NO (-45.8%)	NO
L00: Kitchen	NO (-64.7%)	NO
L00: Storage	N/A	N/A
L00: Ladies Ablution/ Toilet	N/A	N/A
L00: Circulation Area	N/A	N/A
L00: Disabled Toilet	N/A	N/A

Regulation 25A: Consideration of high efficiency alternative energy systems

Were alternative energy systems considered and analysed as part of the design process?	YES
Is evidence of such assessment available as a separate submission?	YES
Are any such measures included in the proposed design?	YES

Technical Data Sheet (Actual vs. Notional Building)

Building Global Parameters

	Actual	Notional
Floor area [m ²]	359.8	359.8
External area [m ²]	1026	1026
Weather	LON	LON
Infiltration [m ³ /hm ² @ 50Pa]	3	3
Average conductance [W/K]	201.27	347.35
Average U-value [W/m ² K]	0.2	0.34
Alpha value* [%]	12.7	10

* Percentage of the building's average heat transfer coefficient which is due to thermal bridging

Building Use

% Area Building Type

Retail/Financial and Professional Services
 Restaurants and Cafes/Drinking Establishments/Takeaways
 Offices and Workshop Businesses
 General Industrial and Special Industrial Groups
 Storage or Distribution
 Hotels
 Residential Institutions: Hospitals and Care Homes
 Residential Institutions: Residential Schools
 Residential Institutions: Universities and Colleges
 Secure Residential Institutions
 Residential Spaces
 Non-residential Institutions: Community/Day Centre
 Non-residential Institutions: Libraries, Museums, and Galleries
 Non-residential Institutions: Education
 Non-residential Institutions: Primary Health Care Building
 Non-residential Institutions: Crown and County Courts
77 General Assembly and Leisure, Night Clubs, and Theatres
 Others: Passenger Terminals
23 Others: Emergency Services
 Others: Miscellaneous 24hr Activities
 Others: Car Parks 24 hrs
 Others: Stand Alone Utility Block

Energy Consumption by End Use [kWh/m²]

	Actual	Notional
Heating	6.68	9.99
Cooling	4.32	3.79
Auxiliary	19.78	19.96
Lighting	5.45	10.89
Hot water	5.85	5.56
Equipment*	15.1	15.1
TOTAL **	42.08	50.19

* Energy used by equipment does not count towards the total for consumption or calculating emissions.

** Total is net of any electrical energy displaced by CHP generators, if applicable.

Energy Production by Technology [kWh/m²]

	Actual	Notional
Photovoltaic systems	46.37	0
Wind turbines	0	0
CHP generators	0	0
Solar thermal systems	0	0
<i>Displaced electricity</i>	<i>46.37</i>	<i>0</i>

Energy & CO₂ Emissions Summary

	Actual	Notional
Heating + cooling demand [MJ/m ²]	118.46	163.11
Primary energy [kWh _{PE} /m ²]	-6.21	74.44
Total emissions [kg/m ²]	-0.28	6.9

HVAC Systems Performance										
System Type	Heat dem MJ/m2	Cool dem MJ/m2	Heat con kWh/m2	Cool con kWh/m2	Aux con kWh/m2	Heat SSEFF	Cool SSEER	Heat gen SEFF	Cool gen SEER	
[ST] Fan coil systems, [HS] ASHP, [HFT] Electricity, [CFT] Electricity										
	Actual	53.9	64.5	6.7	4.3	19.8	2.24	4.15	2.5	5
	Notional	100	63.2	10	3.8	20	2.78	4.63	----	----

Key to terms	
Heat dem [MJ/m2]	= Heating energy demand
Cool dem [MJ/m2]	= Cooling energy demand
Heat con [kWh/m2]	= Heating energy consumption
Cool con [kWh/m2]	= Cooling energy consumption
Aux con [kWh/m2]	= Auxiliary energy consumption
Heat SSEFF	= Heating system seasonal efficiency (for notional building, value depends on activity glazing class)
Cool SSEER	= Cooling system seasonal energy efficiency ratio
Heat gen SSEFF	= Heating generator seasonal efficiency
Cool gen SSEER	= Cooling generator seasonal energy efficiency ratio
ST	= System type
HS	= Heat source
HFT	= Heating fuel type
CFT	= Cooling fuel type